

Requirement Analysis

Solution Requirements

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Project Name	Automated Network Request Management in ServiceNow

1. Functional Requirements

The following functional requirements define the end-to-end operational capabilities of the automated Network Request management solution in ServiceNow, encompassing catalog intake, dynamic UI validations, backend persistence, workflow routing, and notification dispatching.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Catalog Request Intake & Portal Access	• FR-1.1: Enable self-service submission via Service Portal (/sp) under Network Services. • FR-1.2: Capture requester profile information (Requested For, Opened on behalf of, User Name, Phone Number, Email ID, Mobile Number). • FR-1.3: Support multi-file attachment uploads for network diagrams, IP requests, and ID proof.
FR-2	Dynamic Form Logic & Input Validation	• FR-2.1: Enforce mandatory submission rules on critical fields (Type of Connection, Address, Total

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		Amount). • FR-2.2: Implement Catalog UI Policies to dynamically display Existing ID only when Type of Connection is selected as Existing. • FR-2.3: Provide radio-choice selection for Mode of Payment (UPI, Card) with instantaneous validation.
FR-3	Backend Persistence & Data Modeling	• FR-3.1: Automatically map submitted catalog variables into the custom persistence table Database Tables (u_database_table). • FR-3.2: Generate unique auto-numbered record identifiers (e.g., DBT0001001) upon submission. • FR-3.3: Maintain bidirectional link between sc_req_item and u_database_table records.
FR-4	Automated Flow Designer Orchestration	• FR-4.1: Trigger Flow Designer workflow asynchronously upon Service Catalog submission. • FR-4.2: Execute Get Catalog Variables action to parse intake fields into workflow data pills. • FR-4.3: Execute Create Record action targeting Database Tables (u_database_table) with initial state Awaiting Approval.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-5	Governance & Approval Lifecycle	• FR-5.1: Execute Ask For Approval action linked to the Database Tables record targeting designated approvers/managers. • FR-5.2: Evaluate conditional decision branch (If: Ask For Approval -> Approval State is Approved). • FR-5.3: Update Database Tables record state automatically to Approved on True branch, or Rejected on False branch.
FR-6	Automated Email Notifications	• FR-6.1: Dispatch instant submission acknowledgment email to requester containing record reference. • FR-6.2: Dispatch automated notification email upon successful request approval. • FR-6.3: Dispatch rejection notification email with business justification upon request denial.

Table 2: Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description & Implementation Standard
NFR-1	Usability	The Service Portal interface must provide an intuitive single-page form layout adhering to WCAG 2.1 accessibility standards. Catalog UI policies must dynamically present relevant fields without full-page reloads, ensuring zero user ambiguity during submission.
NFR-2	Security & Access Control	All data transmissions must be encrypted via TLS 1.3, with data at rest secured using AES-256. Access Control Lists (ACLs) on Database Tables (u_database_table) restrict record write/modify privileges to authorized administrative roles while requesters retain read-only visibility over their own tickets.
NFR-3	Reliability & Data Integrity	The solution enforces mandatory field integrity on both client-side policies and database-level constraints. Transactional rollback mechanisms prevent orphaned approval records if flow execution fails, ensuring 100% data consistency.
NFR-4	Performance	Catalog item form render time on the Service Portal must remain under 2.0 seconds under normal load. Automated Flow Designer execution (record creation, variable parsing, and email dispatch) must complete within 5 seconds of submission.

NFR No.	Non-Functional Requirement	Description & Implementation Standard
NFR-5	Availability	The application architecture leverages ServiceNow's Advanced High Availability (AHA) infrastructure with active-active cloud clusters, targeting 99.98% operational uptime and zero data loss via redundant real-time database replication.
NFR-6	Scalability	The modular Flow Designer architecture and multi-tenant database indexing must effortlessly scale to support 1,000+ concurrent network submissions per day without requiring structural refactoring or causing thread contention.